SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA1402 **COURSE NAME: Compiler Design for High-Level Programming**

COURSE OUTCOME COVERED

CO3: *Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 60 **Total Marks:60**

Annexure A

MCQ

Code of Conduct:

*I_ **Monika.R(192324281)_**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Monika.R

Criteria	Weightage	Level 4 – Exemplary (50–60 Marks)	Level 3 – Proficient (40–49 Marks)	Level 2 – Developing (30–39 Marks)	Level 1 – Beginning (0–29 Marks)
Number of Correct Answers	100%	to 60 correct answers	40 to 49 correct answers	30 to 39 correct answers	0 to 29 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

Course Faculty

CSA1402- COMPILER DESIGN

CO3-AT3

MULTIPLE CHOICE QUESTIONS

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Submitted To:	Submitted on:
Dr. N. Deepa	12-08-2026

Annexure A

MCQ

Question: 1

A compiler designer wants to attach semantic meaning to grammar productions in order to generate intermediate representations efficiently. Which compiler concept is most suitable?

A) Lexical buffering

B) Associating semantic rules with grammar productions

C) Register allocation

D) Peephole optimization

Answer: B) Associating semantic rules with grammar productions

Explanation: Semantic rules give meaning to grammar productions and help perform translation.

Question: 2

While compiling an arithmetic expression, a compiler internally creates a hierarchical representation showing operators and operands. Which structure is being generated?

A) Stack frame

B) Queue structure

C) Syntax Tree

D) Linked List

Answer: C) Syntax Tree

Explanation: A syntax tree represents operators and operands in a hierarchical structure.

Question: 3

A compiler evaluates attributes from child nodes before computing the parent node value during semantic analysis. Which type of attribute evaluation is taking place?

A) Parent-driven evaluation

B) Synthesized attribute evaluation

C) Inherited attribute evaluation

D) Dynamic attribute evaluation

Answer: B) Synthesized attribute evaluation

Explanation: Synthesized attributes obtain their values from child nodes.

Question: 4

A language processor uses LR parsing to evaluate semantic information during reductions. Which attribute grammar is naturally suitable in this case?

A) Context-free attribute grammar

B) Recursive attribute grammar

C) S-attributed definition

D) Circular attribute grammar

Answer: C) S-attributed definition

Explanation: S-attributed definitions use only synthesized attributes and work naturally with bottom-up LR parsing.

Question: 5

During semantic rule evaluation, a parser requires both parent information and child information while traversing grammar symbols from left to right. Which definition supports this requirement?

A) S-attributed definition

B) Static semantic definition

C) L-attributed definition

D) Ambiguous definition

Answer: C) L-attributed definition

Explanation: L-attributed definitions allow inherited information to flow from the parent or left-side symbols.

Question: 6

A compiler engineer wants child nodes to be processed before the parent node during semantic evaluation of expressions. Which traversal strategy should be selected?

A) Preorder traversal

B) Inorder traversal

C) Postorder traversal

D) Level-order traversal

Answer: C) Postorder traversal

Explanation: Postorder visits children first and the parent node last.

Question: 7

An expression such as (a+b)*c is represented internally without unnecessary grammar details. What does this representation mainly capture?

A) Memory layout

B) Hierarchical program structure

C) Register allocation

D) CPU execution order

Answer: B) Hierarchical program structure

Explanation: A syntax tree shows the hierarchical relationship between operators and operands.

Question: 8

While evaluating semantic rules, information is passed from a parent node toward its child nodes. Which type of attribute is involved?

A) Synthesized attribute

B) Terminal attribute

C) Inherited attribute

D) Dynamic attribute

Answer: C) Inherited attribute

Explanation: Inherited attributes receive information from parent or neighboring nodes.

Question: 9

A compiler rejects the statement int x = "hello"; before execution begins. Which feature of the programming language is responsible for detecting this issue?

A) Token generation

B) Type system

C) Syntax optimization

D) Loop unrolling

Answer: B) Type system

Explanation: The type system detects incompatible data types such as assigning a string to an integer.

Question:10

Consider the statement float value = 25;. Which compiler activity occurs automatically here?

A) Explicit casting

B) Dynamic conversion

C) Implicit type conversion

D) Symbol replacement

Answer: C) Implicit type conversion

Explanation: The integer value is automatically converted to float without explicit casting.

Question: 11

A semantic analyzer verifies whether variables are declared and operations are type-compatible. Which compiler phase performs these tasks?

A) Lexical Analysis

B) Syntax Analysis

C) Semantic Analysis

D) Linking

Answer: C) Semantic Analysis

Explanation: Semantic analysis checks declarations, types, scope, and meaning of statements.

Question: 12

A compiler developer chooses a representation that eliminates unnecessary grammar symbols while preserving expression meaning. Which structure is preferred over a parse tree?

A) Symbol Table

B) Syntax Tree

C) Token Stream

D) Finite Automaton

Answer: B) Syntax Tree

Explanation: A syntax tree removes unnecessary grammar details while preserving the program structure.

Question: 13

During compilation, all information regarding identifiers, scope, and datatype is maintained centrally. Which compiler component performs this responsibility?

A) Code Generator

B) Symbol Table

C) Syntax Parser

D) Optimizer

Answer: B) Symbol Table

Explanation: The symbol table stores information about identifiers, their types, and scopes.

Question: 14

A compiler uses only synthesized information for semantic evaluation of grammar productions. Which type of attribute grammar is being implemented?

A) Dynamic grammar

B) L-attributed grammar

C) S-attributed grammar

D) Context-sensitive grammar

Answer: C) S-attributed grammar

Explanation: S-attributed grammars contain only synthesized attributes.

Question: 15

A recursive descent parser performs semantic translation while expanding productions from the start symbol downward. Which translation approach is being followed?

A) Bottom-up translation

B) Top-down translation

C) Shift-reduce translation

D) Operator precedence translation

Answer: B) Top-down translation

Explanation: Recursive descent parsing starts from the start symbol and proceeds downward.

Question: 16

During shift-reduce parsing, intermediate grammar symbols are temporarily stored before reductions occur. Which data structure supports this operation most effectively?

A) Queue

B) Heap

C) Stack

D) Linked List

Answer: C) Stack

Explanation: Shift-reduce parsers use a stack to store symbols before reduction.

Question: 17

A compiler reports an error when a boolean value is added to an integer variable. What compiler mechanism detected this problem?

A) Syntax checking

B) Type checking

C) Token validation

D) Memory management

Answer: B) Type checking

Explanation: Type checking detects operations performed between incompatible data types.

Question: 18

A programmer declares a variable to store whole numbers without decimal values. Which datatype is most appropriate?

A) Boolean

B) Integer

C) Character

D) String

Answer: B) Integer

Explanation: Integer data types store whole-number values without decimals.

Question: 19

Two array types are considered equivalent because they possess identical internal structures even though their names differ. Which type equivalence principle is applied?

A) Name equivalence

B) Structural equivalence

C) Functional equivalence

D) Dynamic equivalence

Answer: B) Structural equivalence

Explanation: Structural equivalence depends on the internal structure and components of the types.

Question: 20

A compiler implementing LR parsing evaluates semantic rules during reductions. Which evaluation strategy is best suited for this implementation?

A) Top-down evaluation

B) Parallel evaluation

C) Bottom-up evaluation

D) Recursive evaluation

Answer: C) Bottom-up evaluation

Explanation: LR parsing is bottom-up, so semantic evaluation naturally occurs during reductions.

Question: 21

During expression analysis, a compiler stores operators as internal nodes and operands as leaves. Which representation is being used?

A) AVL Tree

B) Heap Tree

C) Syntax Tree

D) B-Tree

Answer: C) Syntax Tree

Explanation: In a syntax tree, operators are internal nodes and operands are generally leaf nodes.

Question: 22

A semantic rule requires information from symbols appearing to the left side of a production. Which attribute grammar supports such evaluation?

A) S-attributed grammar

B) L-attributed grammar

C) Ambiguous grammar

D) Recursive grammar

Answer: B) L-attributed grammar

Explanation: L-attributed grammars allow information to be obtained from symbols on the left.

Question: 23

After semantic analysis, a compiler converts validated source constructs into machine-independent instructions. Which compiler phase performs this task?

A) Lexical Analysis

B) Intermediate Code Generation

C) Linking

D) Parsing

Answer: B) Intermediate Code Generation

Explanation: Intermediate code generation produces machine-independent intermediate instructions.

Question: 24

A programmer intentionally converts a floating-point value into an integer using a language construct. What is this process called?

A) Promotion

B) Reduction

C) Casting

D) Coercion

Answer: C) Casting

Explanation: Casting is an explicit conversion from one data type to another.

Question: 25

While scanning source code, a compiler groups characters into meaningful units such as keywords and identifiers. Which phase performs this operation?

A) Semantic Analysis

B) Syntax Analysis

C) Lexical Analysis

D) Optimization

Answer: C) Lexical Analysis

Explanation: Lexical analysis converts character sequences into tokens such as keywords and identifiers.

Question: 26

A parse tree generated for a program statement explicitly contains all grammar symbols used in derivation. What is the primary characteristic of this structure?

A) Contains machine instructions

B) Represents grammar symbols

C) Removes terminals

D) Stores registers only

Answer: B) Represents grammar symbols

Explanation: A parse tree shows the grammar symbols used during the derivation of the input.

Question: 27

A bottom-up parser starts constructing derivations beginning with identifiers and constants before reaching the start symbol. Which nodes are processed first?

A) Root nodes

B) Internal nodes

C) Leaf nodes

D) Header nodes

Answer: C) Leaf nodes

Explanation: Bottom-up processing begins with the input symbols represented at the leaves.

Question: 28

During assignment checking, a compiler ensures the datatype of the expression matches the datatype of the variable receiving it. Which component performs this verification?

A) Loader

B) Semantic checker

C) Tokenizer

D) Syntax formatter

Answer: B) Semantic checker

Explanation: The semantic checker verifies type compatibility during assignment.

Question: 29

Two variables are treated as equivalent only because they were declared using the same type name. Which equivalence mechanism is being followed?

A) Structural equivalence

B) Dynamic equivalence

C) Name equivalence

D) Functional equivalence

Answer: C) Name equivalence

Explanation: Name equivalence considers types equivalent when they have the same declared type name.

Question: 30

A semantic evaluator computes all child expressions before processing the parent expression. Which traversal technique reflects this behavior?

A) Preorder traversal

B) Inorder traversal

C) Postorder traversal

D) Reverse traversal

Answer: C) Postorder traversal

Explanation: Postorder traversal processes children before their parent.

Question: 31

A compiler designer inserts executable semantic instructions directly inside grammar productions to perform translation tasks. What is this mechanism called?

A) Syntax-directed translation scheme

B) Token generation mechanism

C) Recursive grammar scheme

D) DFA representation

Answer: A) Syntax-directed translation scheme

Explanation: A syntax-directed translation scheme places semantic actions directly within grammar productions.

Question: 32

A parser repeatedly performs shift and reduce operations until the start symbol is produced. Which parsing strategy is being used?

A) Predictive parsing

B) Top-down parsing

C) Bottom-up parsing

D) Recursive parsing

Answer: C) Bottom-up parsing

Explanation: Shift-reduce operations are characteristic of bottom-up parsing.

Question: 33

A strongly typed programming language prevents invalid arithmetic operations between incompatible datatypes. Which language feature enforces this?

A) Type system

B) Code optimizer

C) Symbol allocator

D) Loader

Answer: A) Type system

Explanation: The type system prevents operations that violate datatype compatibility rules.

Question: 34

A programmer creates a collection capable of storing multiple values of the same datatype under one name. Which datatype category does this belong to?

A) Primitive datatype

B) Derived datatype

C) Boolean datatype

D) Character datatype

Answer: B) Derived datatype

Explanation: Arrays and similar collections are derived from basic data types.

Question: 35

In a syntax tree representing an assignment statement, the highest node corresponds to the complete statement meaning. Which element does the root node generally represent?

A) Constant value

B) Identifier name

C) Entire expression or statement

D) Memory address

Answer: C) Entire expression or statement

Explanation: The root represents the overall expression or statement being represented.

Question: 36

A compiler reports an error because a variable is used outside its declared scope. Which compiler phase detects this issue?

A) Syntax Analysis

B) Semantic Analysis

C) Code Optimization

D) Linking

Answer: B) Semantic Analysis

Explanation: Scope and declaration-related errors are detected during semantic analysis.

Question: 37

A compiler automatically converts an integer operand into a floating-point operand before arithmetic evaluation. What type of conversion is this?

A) Narrowing conversion

B) Implicit widening conversion

C) Explicit casting

D) Recursive conversion

Answer: B) Implicit widening conversion

Explanation: Converting an integer to a wider floating-point type automatically is implicit widening.

Question: 38

During attribute evaluation, semantic rules are processed from left to right so that inherited information becomes available when required. Which attribute grammar follows this restriction?

A) Circular grammar

B) L-attributed grammar

C) S-attributed grammar

D) Ambiguous grammar

Answer: B) L-attributed grammar

Explanation: L-attributed grammars restrict inherited attribute dependencies to support left-to-right evaluation.

Question: 39

A compiler uses shift-reduce actions and reduction tables to parse complex programming constructs. Which parser is most likely being used?

A) Predictive parser

B) Recursive descent parser

C) LR parser

D) LL(1) parser

Answer: C) LR parser

Explanation: LR parsers use shift-reduce parsing with parsing tables.

Question: 40

A compiler engineer associates semantic computations directly with grammar rules for translation purposes. To what are these semantic rules attached?

A) Tokens only

B) Grammar productions

C) Machine registers

D) Memory blocks

Answer: B) Grammar productions

Explanation: Semantic rules are associated with grammar productions to define their meaning.

Question: 41

During semantic evaluation, a node computes its value entirely using values obtained from child nodes. Which attribute type is involved?

A) Inherited attribute

B) Dynamic attribute

C) Synthesized attribute

D) External attribute

Answer: C) Synthesized attribute

Explanation: A synthesized attribute is calculated from the attributes of child nodes.

Question: 42

A compiler designer prefers a compact internal representation that removes unnecessary non-terminal symbols from derivations. Which representation is best suited?

A) Parse Tree

B) Syntax Tree

C) NFA

D) Transition Graph

Answer: B) Syntax Tree

Explanation: A syntax tree provides a compact representation by removing unnecessary grammar symbols.

Question: 43

A program contains the statement total = amount + tax; where tax has never been declared. Which compiler phase identifies this error?

A) Lexical Analysis

B) Parsing

C) Semantic Analysis

D) Code Generation

Answer: C) Semantic Analysis

Explanation: Semantic analysis checks whether identifiers such as tax have been properly declared.

Question: 44

A compiler automatically changes the datatype of an operand during expression evaluation to avoid type mismatch. What is this process called?

A) Coercion

B) Parsing

C) Reduction

D) Folding

Answer: A) Coercion

Explanation: Coercion is an automatic conversion of one datatype into another.

Question: 45

Two structures are treated as equivalent because they contain identical field arrangements and datatypes. Which equivalence principle is being applied?

A) Name equivalence

B) Scope equivalence

C) Structural equivalence

D) Dynamic equivalence

Answer: C) Structural equivalence

Explanation: Structural equivalence compares the internal structure and datatypes of the types.

Question: 46

A compiler implementing LR parsing evaluates semantic actions during reduce operations. Which evaluation strategy aligns best with this parser behavior?

A) Predictive evaluation

B) Recursive evaluation

C) Bottom-up evaluation

D) Top-down evaluation

Answer: C) Bottom-up evaluation

Explanation: LR parsing works bottom-up, so semantic actions are naturally evaluated during reductions.

Question: 47

A compiler developer chooses an attribute grammar that can be evaluated naturally during reduce actions without inherited information. Which grammar type is most appropriate?

A) L-attributed grammar

B) S-attributed grammar

C) Ambiguous grammar

D) Circular grammar

Answer: B) S-attributed grammar

Explanation: S-attributed grammars use only synthesized attributes and are suitable for bottom-up evaluation.

Question: 48

In a syntax tree for arithmetic expressions, the terminal values such as variables and constants appear at which location?

A) Internal nodes

B) Root node

C) Leaf nodes

D) Header nodes

Answer: C) Leaf nodes

Explanation: Variables and constants are represented as leaf nodes in a syntax tree.

Question: 49

A programmer explicitly converts a floating-point result into an integer before storing it in a variable. Which operation is being applied?

A) Promotion

B) Casting

C) Translation

D) Coercion

Answer: B) Casting

Explanation: Explicitly converting a value from float to integer is called casting.

Question: 50

After successful syntax checking, a compiler begins verifying meanings, datatypes, and scope rules. Which phase starts next?

A) Lexical Analysis

B) Semantic Analysis

C) Linking

D) Loading

Answer: B) Semantic Analysis

Explanation: Semantic analysis follows syntax analysis and checks meaning, types, and scope.

Question: 51

A programming language verifies datatype compatibility during compilation and sometimes again during execution. What does this indicate about type checking?

A) It occurs only during loading

B) It occurs only during runtime

C) It can occur during compile time and runtime

D) It is unnecessary in modern languages

Answer: C) It can occur during compile time and runtime

Explanation: Type checking may be performed statically at compile time and dynamically at runtime.

Question: 52

A parser constructs derivations beginning from tokens and gradually reduces them until the start symbol is obtained. Which parsing approach is demonstrated?

A) Top-down parsing
B) Predictive parsing
C) Bottom-up parsing
D) Recursive parsing

Answer: C) Bottom-up parsing
Explanation: Bottom-up parsing starts with input tokens and reduces them toward the start symbol.

Question: 53

A compiler internally represents expressions using parent-child relationships between operators and operands instead of linear text order. Which representation style is being followed?

A) Sequential representation
B) Hierarchical representation
C) Tabular representation
D) Circular representation

Answer: B) Hierarchical representation
Explanation: Parent-child relationships represent expressions hierarchically.

Question: 54

A compiler phase determines whether expressions, assignments, and declarations carry valid meaning according to language rules. Which aspect is primarily analyzed?

A) Syntax structure
B) Meaning of constructs
C) Register allocation
D) Instruction scheduling

Answer: B) Meaning of constructs
Explanation: Semantic analysis determines whether program constructs have valid meaning.

Question: 55

Two datatype declarations are accepted as compatible because the language rules consider them logically interchangeable. What does this describe?

A) Operator precedence

B) Variable scope

C) Type equivalence

D) Memory allocation

Answer: C) Type equivalence

Explanation: Type equivalence determines whether two data types are considered compatible according to language rules

Answers:

1. B) Associating semantic rules with grammar productions
2. C) Syntax Tree
3. B) Synthesized attribute evaluation
4. C) S-attributed definition
5. C) L-attributed definition
6. C) Postorder traversal
7. B) Hierarchical program structure
8. C) Inherited attribute
9. B) Type system
10. C) Implicit type conversion
11. C) Semantic Analysis
12. B) Syntax Tree
13. B) Symbol Table
14. C) S-attributed grammar
15. B) Top-down translation
16. C) Stack
17. B) Type checking
18. B) Integer
19. B) Structural equivalence
20. C) Bottom-up evaluation
21. C) Syntax Tree
22. B) L-attributed grammar
23. B) Intermediate Code Generation
24. C) Casting
25. C) Lexical Analysis
26. B) Represents grammar symbols
27. C) Leaf nodes
28. B) Semantic checker
29. C) Name equivalence
30. C) Postorder traversal
31. A) Syntax-directed translation scheme
32. C) Bottom-up parsing
33. A) Type system
34. B) Derived datatype
35. C) Entire expression or statement

36. B) Semantic Analysis
37. B) Implicit widening conversion
38. B) L-attributed grammar
39. C) LR parser
40. B) Grammar productions
41. C) Synthesized attribute
42. B) Syntax Tree
43. C) Semantic Analysis
44. A) Coercion
45. C) Structural equivalence
46. C) Bottom-up evaluation
47. B) S-attributed grammar
48. C) Leaf nodes
49. B) Casting
50. B) Semantic Analysis
51. C) It can occur during compile time and runtime
52. C) Bottom-up parsing
53. B) Hierarchical representation
54. B) Meaning of constructs
55. C) Type equivalence

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (90–100 Marks)	Level 3 – Proficient (75–89 Marks)	Level 2 – Developing (50–74 Marks)	Level 1 – Beginning (0–49 Marks)
Number of Correct Answers	**100%**	**90 to 100 correct answers**	**75 to 89 correct answers**	**50 to 74 correct answers**	**0 to 49 correct answer s**

Criteria	Weight age (%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understandin g	30	Demonstrate s clear and complete understandin g of the concept	Good understandi ng with minor gaps	Basic understandi ng with some misconcepti ons	Poor or incorrect understan ding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/prob lem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completenes s of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and	Understand able with	Somewhat unclear or	Difficult to understan d

Criteria	Weightage (%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
		well-organized answer	minor issues	poorly structured	